Do Earthquakes Exhibit Self-Organized Criticality?

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If earthquakes are phenomena of self-organized criticality (SOC), statistical characteristics of the earthquake time series should be invariant after the sequence of events in an earthquake catalog are randomly rearranged. In this Letter we argue that earthquakes are unlikely phenomena of SOC because our analysis of the Southern California Earthquake Catalog shows that the first-return-time probability $P_M(T)$ is apparently changed after the time series is rearranged. This suggests that the SOC theory should not be used to oppose the efforts of earthquake prediction.

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During the debate on earthquake prediction published on Nature magazine's Web site [1], a controversial and hotly debated topic is whether earthquakes are phenomena of self-organized criticality (SOC). There are two important implications if earthquakes are indeed of SOC: (i) earthquakes can occur randomly anywhere at any time and (ii) an earthquake cannot "know" how large it will become [1]; in other words, earthquakes cannot be predicted [2].

SOC is a theory for large interactive systems and states that, when large disparity exists between the time scales associated with the system driving and response, such systems will evolve naturally toward a critical state so that a minor event can lead to a catastrophe through an avalanche process [3]. Both computer modeling and laboratory measurement of sandpile experiments—the most noticeable paradigm of SOC—have revealed the power-law frequency distribution for the sizes of events. Power-law distribution has been considered as the signature characteristic for SOC phenomena and has been the essential reason [4] to consider earthquake as a SOC phenomenon because the power law resembles the Gutenberg-Richter (GR) law [5]. The conjecture of an earthquake being of SOC leads to an adverse suggestion to efforts of earthquake prediction because SOC means that it is impossible to predict where and when an earthquake will occur and how strong it will become [1]. However, the analysis of Southern California Seismographic Network (SCSN) Catalog Data [6] reveals that the first-return-time probability, $P_M(T)$ (defined as the probability that if an earthquake of magnitude equal to or greater than M occurs at time T_0, another quake of magnitude equal to or greater than M will occur at $T_0 + T$), exhibits power-law distribution [7], which is fundamentally inconsistent with SOC models: computer modeling of sandpile experiments suggests an exponential distribution. This certainly casts doubt on the applicability of SOC to earthquake prediction.

We analyzed 19 years (1984–2002) of SCSN catalog data to demonstrate that earthquakes are not unpredictable. Our main focus is on the system-wide (spanning 29°–40° N in latitude and 113°–123° W in longitude) first-return-time probability. We intentionally ignored the effect of spatial distribution so that we could examine if the system-wide $P_M(T)$ was similar to that obtained in the analysis by dividing the SCSN region into a number of smaller cells [7]. We calculated $P_M(T)$ for all earthquakes of magnitude $M \geq 2$ regardless where the epicenters were located. As shown in Fig. 1, the power-law regime spans for 3–4 decades, depending on the earthquake magnitude, and the high end of the power-law regime is $T_C \sim 250 \times 10^M$. This T_C could be an important time scale to characterize a certain physical mechanism involved in earthquakes. If this is the case, it implies that SOC may not be applicable to describe earthquakes because SOC is scale independent [8]. It should be noted that the power law resembles the Omori law [9]

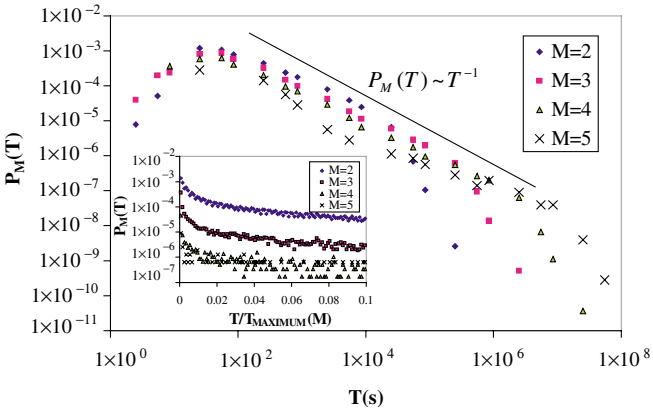


FIG. 1 (color online). First-return-time distribution for actual data of the earthquake catalog. The Omori law is shown as the solid line. The inset shows the distribution for small Ts in semilog coordinates—a straight line indicates an exponential distribution. For Figs. 1–4 each data point represents an average of $P_M(T)$ in a bin. In log-log plots a decade of T is divided into three bins. In semilog plots uniform bin size is used. $T_{\text{maximum}}(M)$ is the largest time separation for events of magnitude $m \geq M$.

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$[P_M(T) \sim T^{-1}]$ for aftershocks. All these features are similar to what has been reported in the analysis carried out by dividing the SCSN region into smaller cells [7]. This result is consistent with the study of the waiting-time (time difference between two successive events) distribution of x-ray solar flares: the power-law distributions for the system-wide region (anywhere on the sun's surface) and an active subregion are almost identical [10].

One implication of earthquakes being of SOC is that an earthquake does not know how large it will become, or, in other words, the magnitude of an earthquake is completely random for a given quake (regardless of the distribution of the stress field). We examine this conjecture by calculating $P_M(T)$ from a rearranged time series: the new time series consists of earthquakes occurring at the same instants as in the actual catalog, but all quakes of magnitude $M \geq 2$ are rearranged in such a way that the magnitudes are drawn randomly from a distribution consistent with the GR law (the SCSN catalog data follow the GR law with $b = 1$; therefore when drawing a random magnitude we use $b = 1$ so that the distribution of the numerically generated earthquake catalog is statistically the same as the actual catalog). If earthquakes did not know how large they would be, $P_M(T)$ calculated from the rearranged catalog would remain the same as that for the original catalog. However, Fig. 2 shows that this is not the case: the power law holds for a narrower range for earthquakes $M \geq 3$ (and the power-law regime disappears for shorter T); e.g., for the case of $M = 4$, the original data give a power law for about 4 decades of T, but the rearranged data follow the power law for merely 2 decades. The difference caused by rearranging the data suggests that earthquakes do not happen with completely random magnitudes, but rather, with a certain degree of randomness, know how large they will become.

Now we go even further and reshuffle the catalog for all the events of $M \geq 2$ [where rearranging events of $M < 2$ is irrelevant because we calculate only $P_M(T)$ for $M \geq 2$]: in addition to assigning an earthquake a random magnitude $M \geq 2$ by drawing it from a distribution consistent with the GR law (again, $b = 1$), we rearrange the sequence of $M \geq 2$ events by drawing the occurrence time from a uniform distribution. For this completely rearranged catalog, as shown in Fig. 3, the power law for $P_M(T)$ does not exist anymore, even for light earthquakes ($M = 2$). This exercise suggests that earthquakes do not happen in a totally random way, but are constrained by the right temporal statistics, e.g., $P_M(T)$.

Why does the original catalog possess a power law for $P_M(T)$ for a longer range, and the magnitude-rearranged catalog still has a power-law regime but lasts for a shorter range, and the completely rearranged catalog does not have a power law at all? We believe that this has to do with the temporal clustering of earthquakes, i.e., foreshocks [11,12] and aftershocks [9] before and after mainshocks, and clustering of clusters, i.e., clustering of mainshocks themselves [13]. By randomly reassigning magnitudes to $M \geq 2$ events, temporally correlated events of $M \geq 3$ within the same clusters become less correlated (but still not uncorrelated) because they are redistributed in a way of forcing them to be so (however, for this case, the clustering of clusters remains unaltered). For example, two events of $M \geq 3$ that are separated by less than ten $2 \leq M < 3$ events in the original catalog will likely become separated (on average) by ten such events because the rearranged catalog is still constrained by the GR law. That is, $M \geq 3$ events within a cluster will become less correlated, but the correlation among clusters is not affected. For the completely rearranged (both magnitude and occurrence time) catalog both the clustering within clusters and the clustering of clusters disappear completely because all events are randomly reassigned to the occurrence instants which are drawn from a uniform distribution.

We replotted Fig. 3 with a linear-log scale, as shown in Fig. 4. It clearly demonstrates that $P_M(T)$ obeys an exponential law (at least for not very big time separations; for large separations the data points are so scattered that it is difficult to make a judgment on what distribution they follow). This is exactly what it should be because uncorrelated events are expected to give rise to an exponential distribution [10,14].

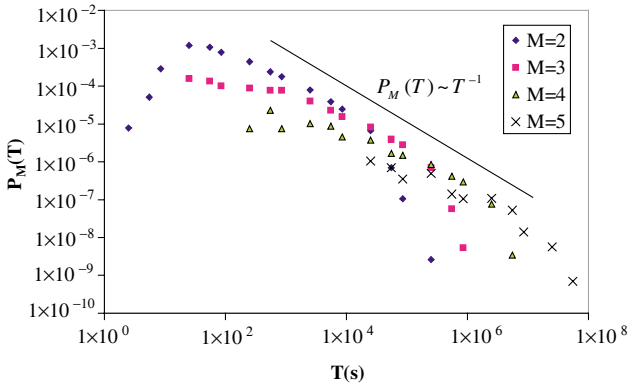


FIG. 2 (color online). Same as Fig. 1 but for magnitude-rearranged data. The Omori law is shown as the solid line.

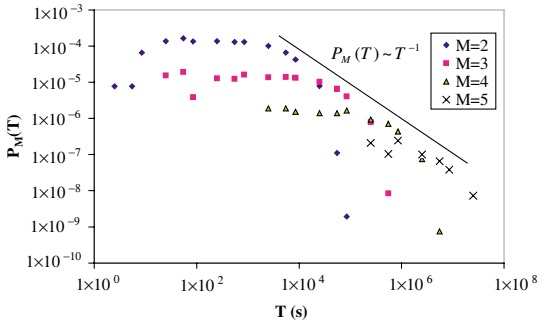


FIG. 3 (color online). Same as Fig. 1 but for totally rearranged (both magnitude and time of occurrence) data. The Omori law is shown as the solid line.

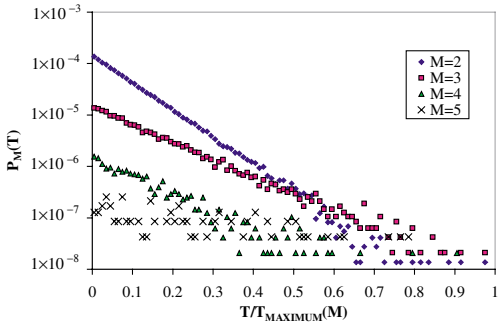


FIG. 4 (color online). Same as Fig. 3 but with log-linear scale. The straight-line effect indicates an exponential distribution. $T_{\text{maximum}}(M)$ is the largest time separation for events of magnitude $m \geq M$.

So far we have demonstrated that the actual earthquake data possess a power-law distribution [for $P_M(T)$], while totally rearranged data obey an exponential distribution. Ironically SOC systems also possess an exponential distribution for first-return-time probability [10,14]; it follows then that it is possible that the totally rearranged earthquake catalog can have a similar temporal correlation as SOC systems, but it is highly unlikely that the actual earthquake catalog will be of any temporal characteristics of SOC.

There is a possibility that the exponential distribution of first-return-time probability for a SOC system could be masked by a power-law distribution either when the experimental resolution (earthquake magnitude for the present study) lies within the power-law regime for the size distribution or when the system is externally driven in a correlated way [14]. However, both sandpile [14] and spring-block [15] numerical experiments indicate that the power-law regime for the first-return-time probability is quite narrow and spans for just 1 decade (at most up to 2 decades for the largest "avalanches" of sizes of 10% of the biggest events possible allowed by the system), but for the earthquake catalog the power-law regime extends 3–4 decades (for events of sizes, in terms of energy release, smaller than 1% of the biggest events). Therefore it is clear that the earthquakes' power-law regime for the first-return-time probability is genuine rather than being an exponential regime masked by a narrow power-law distribution. The second possible explanation is also not applicable: earthquakes are not driven by a correlated stress field because there is no evidence for long-range correlations of the stress field before large earthquakes [1]. Also, a correlated driving can produce a very narrow power-law regime at the high-end tail [14], but the earthquake data display a quite broad one at much shorter time differences.

We believe that the power law for the first-return-time probability exhibited in SOC systems is attributable to the cumulative effect discussed as follows. Recall that $P_M(T)$ is defined for events of magnitude *equal to or greater than* M. Denote the first-return-time probability for events of magnitude *exactly equal to* m is $F_m(T)$, which is related to $P_M(T)$ by $P_M(T) = \int_M^\infty F_m(T)p(m)dm$, where $p(m)$ is the probability density function for size distribution, e.g., the GR law. If we assume that in SOC systems $F_m(T)$ is exponential, i.e., $F_m(T) = \exp(-T/\tau_m)/\tau_m$, where t_m is event-size dependent and without losing generality and further assume $\tau_m = c_1 10^m$ (we will examine these two assumptions in a separate work), with a power-law distribution for $p(m)$, $p(m) = c_2 10^{-m}$, we can easily obtain

$$P_M(T) = \frac{1}{c_1 10^M T_*^2}[1 - T_* \exp(-T_*) - \exp(-T_*)], \quad (1)$$

where $T_* = T/c_1 10^M$. This result is plotted in Fig. 5. It clearly indicates that for large T_* $P_M(T)$ falls off as a power law with the exponent being equal to -2, quite different from that for the Omori law. In some known SOC systems $P_M(T)$ falls off as a -2 power law [15], while in some others it falls off even faster [14]. Another interesting point about Eq. (1), shown in the inset of Fig. 5, is that at small T_* $P_M(T)$ is exponential. This simple analysis suggests that SOC systems can allow both an exponential distribution for the first-return-time probability at small T_* and a power-law distribution at large T_* with a falloff rate that can be much faster than that of the Omori law. These features of Eq. (1) are consistent with the findings reported in [14] (see Figs. 1 and 2 of [14]). The reader is reminded that the exact value of the exponent in the power law is dependent on the form of size distribution, $p(m)$. However, for a SOC system that exhibits GR law consistent power-law size distribution, it is reasonable to expect that the exponent should not be substantially different from -2.

Having said that SOC has an exponential first-return-time probability (at least for not very large T), it is appropriate to add a few words to elaborate why a necessary condition for an earthquake to be a SOC system is the invariance of first-return-time probability after all earthquakes are rearranged, at least at small and intermediate time separations. For a time series of events

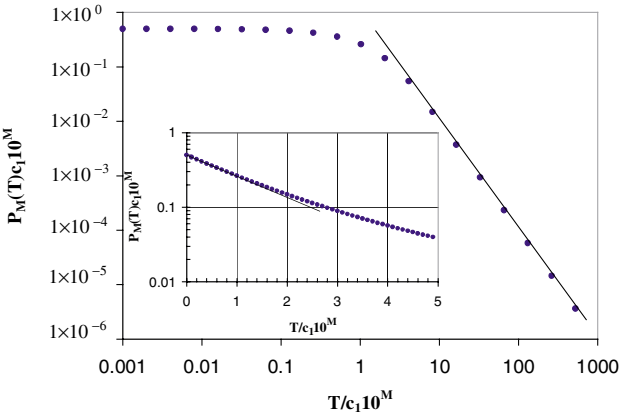


FIG. 5 (color online). An analysis showing how SOC systems can have a power-law distribution for the first-return-time distribution. The falloff rate of the power law is "-2".

generated by a SOC system, its Poisson distribution of first-return-time probability remains Poisson if the time series is mixed up and reordered in a way that is completely random. Therefore a direct test to determine whether a system is of SOC is to examine if its first-return-time probability is altered after the data are rearranged. Worthy of mentioning is the original SCSN catalog does not have an exponential distribution even at small T (shown in the inset of Fig. 1), suggesting again that an earthquake is not a SOC system.

In addition to the invariability of $P_M(T)$ to rearranging the sequence of events in a SOC system, there are two other criteria that help to determine if a system is SOC. The first criterion is that a SOC system has an exponential distribution for $P_M(T)$, at least for small time separations, even for events of sizes above given threshold. This criterion can be practically useful because it means that for events measured with a certain detection limit an exponential distribution can exist at small times for SOC systems; therefore the lack of an exponential distribution indicates the system may not be of SOC. The other criterion is that the falloff rate is apparently faster than that of the Omori law (i.e., a power law with exponent ≈ 1) if a power-law regime indeed exists.

Diffusion entropy analysis of the SCSN catalog data reveals that the Shannon entropy of a diffusion process defined by the seismic sequence is definitely different from a Poisson process, and, furthermore, both the intracluster swarms and intercluster distances exhibit power-law distribution [16]. This is consistent with the present findings. Also, our findings that earthquakes are not SOC phenomena are in agreement with an analysis of the declustered earthquake catalog which demonstrates that main shocks are not scale independent [8].

Before closing, we make a comment about the predictability and the randomness in a dynamic system. A previous work has demonstrated that if a system is deterministic the evolution of the system will be somewhat predictable regardless whether the system is SOC or not [17]. We believe the reason why some SOC systems are not predicable is that these systems are overwhelmed by the randomness constantly imposed on them, e.g., in the sandpile cellular automaton reported in [3]. In our view, the randomness that existed in dynamic systems does not necessarily prohibit predictability. When the evolution of a system is not completely driven by a random force, a certain degree of predictability will emerge. A typical example is weather prediction. Even though at any given location the instantaneous wind speed and air temperature are random variables, spatially and temporally averaged wind speed and air temperature can be predicted with a certain confidence. The reason is simple: many of the driving forces that govern the air motion are deterministic, e.g., solar radiation, earth rotation, and terrain variations.

In summary, in the context of earthquake prediction, our analyses indicate that earthquakes are not SOC phenomena; therefore further research on earthquake prediction should not be discouraged or opposed by the argument that earthquakes are of SOC and as a result unpredictable. Further development of a sound physics-based theory of the precursory process [1], other than the SOC-type theory that is not capable of revealing details of mechanisms of earthquakes, should be pursued to understand earthquakes and to make predictions more reliable than the "black box" pattern of recognition techniques. New technology such as synthetic aperture radars interferometry [18] and a dense network of the global positioning system [19] will provide useful information with unprecedented precision and resolution that will definitely improve the capability of monitoring and predicting earthquakes, just as the GOES satellite and Doppler radar observations have greatly improved daily weather forecasting.

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